# JOSEPH SRUJAN EDA

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## PROFESSIONAL SUMMARY

Embedded Software Engineer with ~3+ years of experience in **Embedded Linux**, **BSP Development**, **Device Drivers**, and **RTOS/Bare-Metal Programming**. Proven track record in board bring-up, Linux kernel tuning, Yocto layer customization, and protocol implementation (UART, I2C, SPI, Modbus, TCP/IP). Experienced in developing high-availability failover mechanisms, industrial IoT gateways, and multi-threaded C applications for critical applications.

## TECHNICAL SKILLS

- **Programming Languages:** C, C++, Data Structures, Bare-Metal Programming
- **Operating Systems & RTOS:** Embedded Linux, FreeRTOS, Linux Kernel, Linux System Programming
- **Protocols & Interfaces:** UART, I2C, SPI, USB, Modbus RTU/TCP, MQTT, TCP/IP
- **BSP & Build Systems:** Board Bring-up, Bootloader (U-Boot) Porting, Yocto Project, Linux Kernel Bring-up
- **Device Drivers:** I2C, SPI, UART, USB Driver Development & Character Device Drivers
- **Microcontrollers & SoCs:** AM335x, ESP32, LPC2148, STM32, Arduino Uno
- **Tools & Debugging:** GDB, Oscilloscopes, Logic Analyzers, Git, Makefile, OpenOCD

## PROFESSIONAL EXPERIENCE

**Kellton Tech** | *Embedded Developer*

*April 2024 – Present* | *Hyderabad, India*

- **Linux Kernel & Device Driver Development:** Developed and integrated custom Linux device drivers for I2C and SPI peripherals on AM335x platforms.
- **BSP & Board Bring-up:** Executed board bring-up procedures, custom bootloader (U-Boot) porting, and kernel configuration using Yocto layers.
- **Industrial IoT Gateway Deployment:** Installed, configured, and commissioned robust industrial IoT gateway panels for field operations (Oil India site), enabling real-time sensor data streaming over Modbus and TCP/IP.
- **Multi-Threaded C Applications:** Engineered high-performance, multi-threaded C applications for gateway communications using POSIX threads, IPC mechanisms, and socket programming.
- **System Stability & Drivers:** Implemented hardware Watchdog timers and low-level diagnostic routines to ensure maximum uptime in harsh field environments.